Calculus I Lesson 11

1) $f(x) = \dfrac{1}{x-2}$

a) $\lim_{x \to 2^-} f(x) = \underline{\quad ? \quad}$

b) $\lim_{x \to 2^+} f(x) = \underline{\quad ? \quad}$

c) $\lim_{x \to 2} f(x) = \underline{\quad ? \quad}$

2) $f(x) = \dfrac{3}{x^2 - 4}$

a) $\lim_{x \to 2^-} f(x) =$ ___?___

b) $\lim_{x \to 2^+} f(x) =$ ___?___

c) $\lim_{x \to 2} f(x) =$ ___?___

d) $\lim_{x \to -2^-} f(x) =$ ___?___

e) $\lim_{x \to -2^+} f(x) =$ ___?___

f) $\lim_{x \to -2} f(x) =$ ___?___

3) $f(x) = \dfrac{4}{(x-1)^2}$

a) $\lim\limits_{x \to 1^-} f(x) =$ ___?___

b) $\lim\limits_{x \to 1^+} f(x) =$ ___?___

c) $\lim\limits_{x \to 1} f(x) =$ ___?___

4) $f(x) = \left(2 + \dfrac{3}{x-2}\right)$

a) $\lim_{x \to 2^-} f(x) = \underline{\quad ? \quad}$

b) $\lim_{x \to 2^+} f(x) = \underline{\quad ? \quad}$

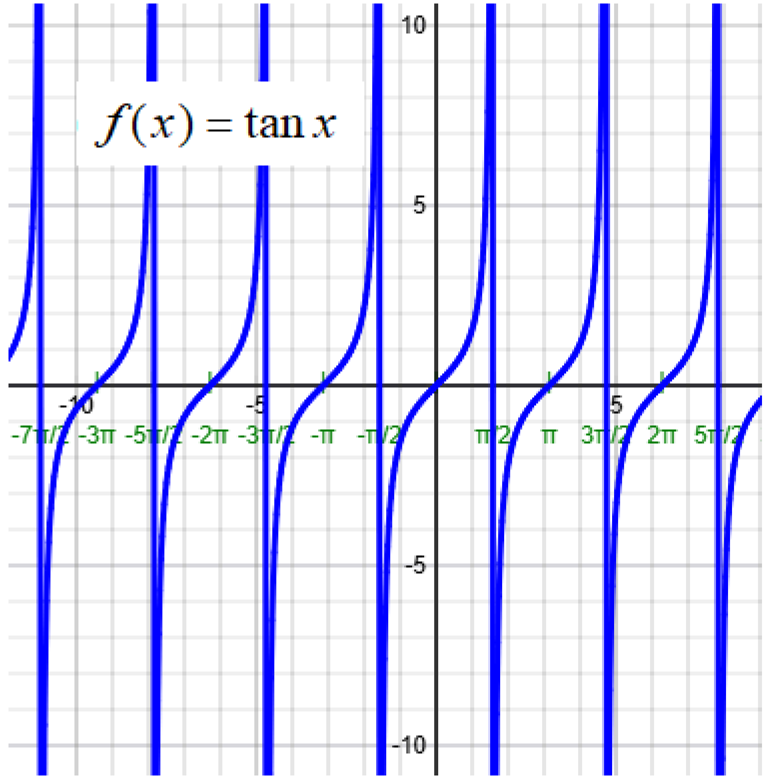
c) $\lim_{x \to 2} f(x) = \underline{\quad ? \quad}$

5) $f(x) = \tan x$

a) $\lim\limits_{x \to \pi/2^-} f(x) =$ ___?___

b) $\lim\limits_{x \to \pi/2^+} f(x) =$ ___?___

c) $\lim\limits_{x \to \pi/2} f(x) =$ ___?___



6) $f(x) = \cot x$

a) $\lim\limits_{x \to \pi^-} f(x) =$ ___?___

b) $\lim\limits_{x \to \pi^+} f(x) =$ ___?___

c) $\lim\limits_{x \to \pi/2} f(x) =$ ___?___